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APPLICATION OF PROCESS MINING TO MEDICAL BILLING USING L* LIFE CYCLE MODEL

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Abstract--- Medical billing is a long chain of processes that cut across different sectors and institutions, evolving into a multimillion-dollar industry. Despite the emergence of institutional involvement in the practice, the parties involved face many challenges that hinder the successful management of the diverse process. Process mining uses computer-based tools and techniques to analyze business processes and ensure best practices for improving business operations as it has a track record of success in improving business processes and procedures. As research on process driving problem solving continues to grow, there is a need to start applying the tools and techniques to specific domain areas and bridge the gap between research, development and application. We demonstrated how process mining techniques can be applied to medical billing using the L* life cycle model from a structured perspective. We developed a handmade comprehensive medical billing process flow using domain knowledge reference and established some time perspective Key Performance Indicators (KPIs) of the billing process. We then explored the hospital billing event log generated from an enterprise resource planning (ERP) system to see how the "as-is" from the event log deviate from the generally known knowledge-based (handmade) model. Taking each trace of the event log as patient visit, we discover that about 63% of all the visit were billed and also about 3% billed got regretted. The event log used in this research does not capture the complete activity flow for a real-world medical billing system. Hence, we were not able to directly compare the process map with the model and give precise operational support

Keywords: process mining, medical billing, L* life cycle

I. INTRODUCTION

Medical billing is a long chain of processes that cut across different sectors and institutions, evolving into a multimillion-dollar industry. Despite the emergence of institutional involvement in the practice, the parties involved still face many challenges that hinder the successful management of the diverse process. Medical billing involves three key stakeholders: the consumer of the healthcare service called patients, healthcare service providers, and the person or institution responsible for the service financial responsibility called the Payer. There have been many challenges in the medical billing process and rising concerns among patients, providers, and payers. Patients who pay high deductible fees and premiums to cover the cost of their care still faced issues such as surprise medical billing

Providers are facing more issues relating to high uncompensated care costs than ever. InstaMed in [1] showed that 4,926 national hospitals in the united states absorbed up to $42.8 billion in uncompensated care, while 5,370 community hospitals recorded a total of $121 billion in 2014. Hospital of all kinds in the United States has provided over $702 billion in uncompensated care since 2000 [2]. Underpayment through Medicaid and Medicare reached $51billion in 2014, and a study in [1] showed that over 30 cents of every dollar in the billing process get wasted on inefficient administrative processes.

This research introduced the "process-thinking" paradigm from a business process management perspective to improving medical billing and reimbursement. "Process-thinking" is not a new phenomenon. It can be traced back to decades of research work by early researchers in diverse fields, such as the great economist Adams Smith and the novel engineer Frederick Taylor [3]. However, it has gained popularity in today's business world, where business process management and data mining are fast becoming the new way of business, with a track record of transformation.

Process mining focus on exploring event log from process-aware information system to generate insight and knowledge about the execution of the process, discovery process model, check the conformance of the event log process with reality, detect, predict and recommend best practices. This paper does not intend to describe process mining in detail but to x-ray the medical billing process from a process mining perspective using (L*) life cycle model proposed by Val der Aalst in [4].

A. *Medical Billing Process*

Researchers have been investigating the challenges in billing management and making various proposals for improvement in this knowledge area over the last decades. Authors in [5] explored the effects of EHRs on the billing and coding processes of vital information for revenue collection while authors in [6] analyze the relationship between hospitals' performance in managing their revenue from billing and their ability to grow equity capital using hospital-level fixed effects regression analysis. Authors in [7] proposed a big data and advanced analytics approach to solving "revenue leakage" to help struggling providers keep their heads above waters in their revenue management strategy while the authors in [8] enumerate logical perspective to effective revenue cycle and billing process management in the era of electronic health. The authors in [9] evaluated the need for medical billing processes and revenue management that focuses on the patient flow process. As stated by [10], information technology as an enabler for healthcare administrative activities and process mining remains unexplored as research in information technology only emphasizes the clinical aspect of healthcare delivery.

There has been little research work focusing on medical billing from a process flow and management perspective, hence the need to focus on progressive process management in medical billing from a business approach using management strategies and bodies of knowledge embodied in business process management and process mining which is the focus of this research.

B. *Process Mining*

Business process management is a comprehensive approach that eases business operations and transformation using a combined knowledge from computer science and management sciences [3], [11]. Business processes are composed of activities that are related to each other and transformed to achieve specific results. Process mining, a subset of business process management (BPM), entails extracting information from business event logs [7] of process-aware information systems for process management activities.

Val der Aalst in [12] enumerated four (4) perspectives to process mining (control flow; performance, conformance and organization) which are most commonly used in literature and involves:

- Control flow perspective: examine the execution flow commonly called workflow order of event that makes up a complete cycle of a business operation using business process management tools such as Petri net.
- Performance perspective: is a time-based analysis that examines event logs using time-dependent metrics such as bottleneck, idle time between activities and resources.
- Conformance perspective: compare the known process flow of the event to those generated from the event log to check the deviation between the event log model and reality.
- Organization Perspective: examine the relationship among personnel and other resources that carry out business process activity to check productivity.

II. RESEARCH METHOD

The method used in this research is that outlined and used by the author in [4] which is shown in Fig. 1. The method consists of 5 stages, namely: stage 0: plan and justifies, stage 1: extract, stage 2: create a control-flow model and connect event log, stage 3: create integrated process model and stage 4: operational support.

A. *Stage 0: Plan and Justify*

This stage enumerates the plans and justification for a proposed process mining project using three approaches namely data-driven, question-driven, and goal-driven.

- Data-driven: involves exploratory analysis of event data without any specific goal or questions using the event log. Although there are no particular questions from the event data, the end product can provide meaningful insight to support business decisions.
- Question Driven: involves analysis of event log in other to answer specific business questions such as why, how, what, when e.t.c. events occur in the process under study.
- Goal-Driven involved analyzing event data based on predefined business key performance indicators (KPIs) or objective functions.

In what follows, we used the hospital billing event log published by [13] as the data to check the flow sequence of all the activities as contained in the L* model. Then, a question-driven approach is used to examines the duration of activities and possible abnormalities in the event log. Knowledge-based medical billing process is used to generate a handmade event log, and prompt payment law is used to generate KPI for specific times for claim submission and payment.

B. *Stage 1: Extract*

During stage 1, we extracted event logs, handmade models, questions from process-aware information systems, domain experts, and other related stakeholders using the information elicitation process based on the plan and justification from stage 0. Objectives are formulated and expressed in terms of key performance indicators in a goal-driven approach.

- Historical Data

The event logs were extracted from the financial module of an ERP system of a regional hospital and contain billing information of the services provided by the hospital. The event log includes a total of 100,000 traces and 18 activities. Each trace has series of activities executed to bill a medical service using the ERP system. The author anonymized activities and events due to the sensitivity of medical data, but timestamps were left unaltered.

- Handmade Models

According to [14] medical billing process start with a patient visit to their care provider. Necessary patient data are captured or updated to prepare patients for an appointment with a medical practitioner, care or treatment. The information capture is pertinent in the billing process; hence, accurate information is necessary during registration. Provider office verifies the patient insurance policy to underscore what the Payer and the patient will cover as part of the care cost. Then, the patient sees the provider who records the patient for diagnosis, treatment or procedure. Detailed information of the care service render is captured for proper coding and billing process. Medical transcription and coding of the documented patient encounter with physicians is transmitted into appropriate billing format and it is assigned medical billing code base on International

Classification of Disease (ICD) and Current Procedure Terminology (CPT) codes used to generate a superbill. Medical service rendered to patients is assigned a fixed cost base of medical transcription and coding and transmitted for submission to Payer either manually or via EDI (Electronic Data Interchange).

The Payer on the other end evaluate the claim for validity and compliance and mark the claim as either accepted, rejected, or denied. When a claim is accepted, the Payer estimates the reimbursement amount, adjustment amount and generates an Explanation of Benefit (EOB) or Electronic Remittance Advice (ERA) transmitted to the provider. The provider creates a payment statement, bills the patient off the balance if applicable, and follows up with patient payment collection. Rejected claims are returned to providers to make necessary corrections before retransmitting to the Payer, while denied claims are claims the Payer is not liable to pay. In most cases, this type of claim is the responsibility of the patient to pay the provider the standard deductible. Fig 2 shows a simplified process flows that representation current medical billing process. The process flow starts with patient visit to a provider at a facility and end with collection from payer and/or patient depending on contract between the payer and the provider.

- Objectives and KPIs

As stated by [15], a clean claims is that which has all necessary fields from a CMS-1500 or CMS 1450 claim form fielded correctly is shown in table 1. A provider should submits a claim to the insurer on or before 90 days after the date of service. The Payer validates the claim and submit payment for the accepted claim, reject or deny the claim within 30 days. The provider receives feedback on rejected or denied claims, make corrections and resubmit the claim to the Payer. The Payer validates and submits payment for rejected or denied claims within 30days. The above sequence of activities is shown in Fig. 3 using the swimlane activity diagram. Based on prompt payment law, for the worst-case scenario, it takes 120 days (90 days submission and 30 days payment) to generate and pay accepted claims and 170 days for rejected or objected claims that are payable.

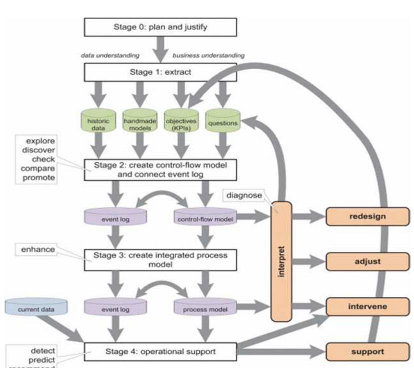

Fig 1. L* cycle model of process mining [4]

TABLE 1. MEDICAL BILLING PROMPT PAYMENT LAW [15]

Provider submit claim	90 days
Payer process claim and submit payment	30 days
Payer rejected or denied the claim	30 days
Provider resubmit rejected or denied claim	20 days
Payer submit payment for previously rejected or denied claim	30 days

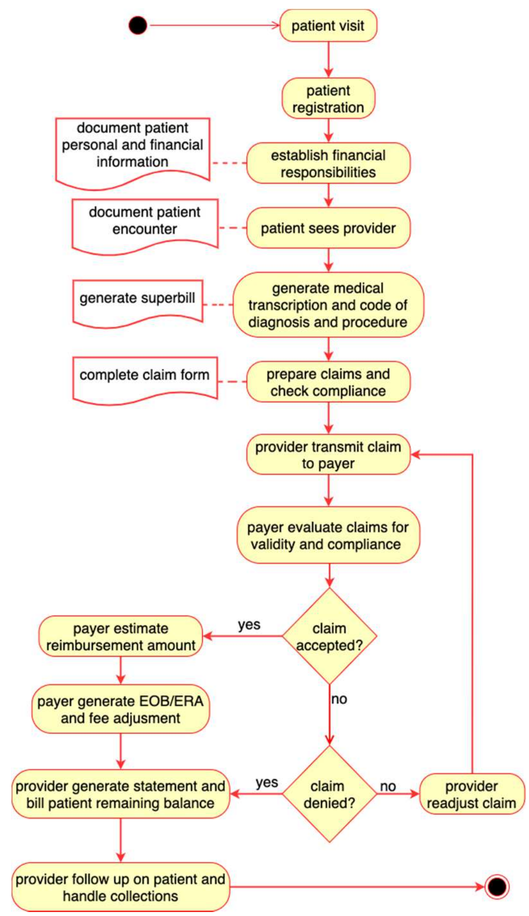

Fig 2. Handmade model of medical billing process flow chart

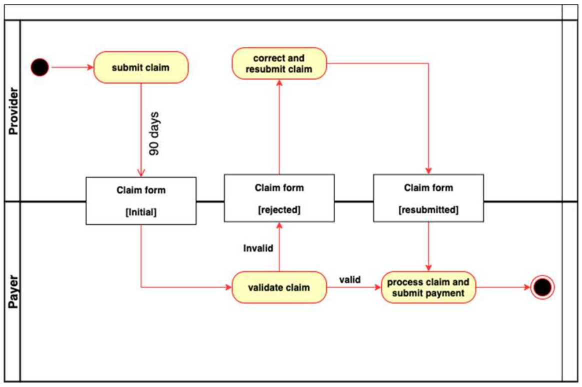

Fig 3. Swimlane activity diagram of claim form transmission between provider and insurer according to prompt payment law

C. Stage 2: Create Control-Flow Model and Connect Event Log

This stage involved ggenerating “de facto” control flow of the event log processes using process mining algorithms, such as fuzzy miner, heuristic mining, alpha algorithm, genetic algorithm etc. Each event in the event log corresponding to an activity in the process model. In cases where a predefined process model exists, perform a conformance check to see if it conforms or deviates from the event log process. The outcome of this stage can inform business decisions and provide answers to questions.

In this research, we generated two control-flow models shown in Fig. 4 and Fig 5; a direct follow graph with 11 of the 18 activities based on their relative frequency and a C-net representing all the activities from the event log using the heuristic algorithm. The discovery process shows a significant deviation from the handmade medical billing process model. Some activities from the handmade models are not present in the event log and vice versa; this might be due to the technical terms used in the ERP system that generates the event log.

D. Stage 3: Create Integrated Process Model

In this stage, enhance the event and process models by adding additional perspectives such as case and organization perspectives and merge the perspectives. The result of this stage is a multipurpose process model that helps to understand better the as-is process, such as identifying bottlenecks.

E. Stage 4: Operational Support

This stage consists of three operational activities which are detect, predict, and recommend. These activities are meant to support business process optimization. The input of this stage is the current business data generated from the event log, also called "pre-mortem data". The output from this stage helps to detect and anomalies (business exception) and to recommend efficient/probable solutions that satisfies or meets the business objectives.

III. RESULTS AND DISCUSSION

The information in table 2 shows statistical representation of each activity in the event log ordered by relative frequency of event. Each trace in the event log started with NEW. The NEW activity in the event log is synonymous to initiating a billing process after patient encounter with provider using the information from the superbill. The activity order is generated using Disco version 2.14.0. Activity with higher frequency (i.e. relative frequency above 10%) are NEW, FIN, RELEASE, CODE OK, and BILLED and CHANGE DIAGN.

Fig. 4 represent the control flow of the event logs generated using ProM 6.10. All activities start from NEW, then split into either CHANGE DIAGN, CHANGE END or JOIN-PAT, and some activity goes from NEW to DELETE or FIN. Activities from JOINT-PAT goes directly to the end while CHANGE DIAGN and CHANGE END goes to DELETE or FIN. Activity from FIN goes to RELEASE which link to coding information of the billing represented as (CODE NOK, CODE OK, CODE-ERROR, and ZDBC_BEHAN). When coding information is not correct (i.e. CODE NOK), the activity flows enter EMPTY and then terminate. When coding information is correct (i.e. CODE OK), it goes to the next state which is either BILLED or REOPEN. CODE-ERROR and ZDBC_BEHAN also goes to BILLED, then to STORENO then REJECT. REJECT goes to SET-STATUS or MANUAL then to the end.

We generated a C-net by filtering out the activities with lower frequency shown in Fig. 5. The C-net comprises of 11 activities which are NEW, CHANGE DIAGN, FIN, DELETE, RELEASE, CODE OK, CODE NOK, BILLED, REOPEN, SATORNO and REJECT. Out of the 100,000 visits, 63, 878 were billed of which 2,002 got rejected for one reason or the other, and 2,379 reopened, which implied that sixty-three percent (63%) of all the visit were not billed, and about three percent (3%) that were billed got rejected.

We introduce time perspective to event log in Fig. 6 to analysis the duration between activities using Disco fuzzy miner algorithm. There is a significant concentration between CODE OK, BILLED, CHANGE DIAGN and FIN, as shown in Fig 6 with duration of 5.8k years (2117 days) and 8.3k years (3030 days). We filter out these activities (CODE OK, BILLED, FIN and CHANGE DIAGNOSIS) with the highest duration for further analysis shown in Fig. 7. All the event goes through FIN, which represents finance; medical coding was done with returning activities about Code status (e.g. CODE OK). Although there is significant activity concentration at these event instances of the event log, this is not enough to affirm whether or not the concentration of the activity results in a bottleneck, because the event log contains random samples of event logs over three years, and also not a complete representation of the overall event logs in that space of time. We recommend that some of the processes that correspond to these activities be examine closely and automated where possible.

TABLE 2. EVENT LOG ACTIVITY ORDERED BY THE RELATIVE FREQUENCY

Activity	Frequency of Event	Relative Frequency of Event
NEW	101,289	22.44%
FIN	74,738	16.58%
RELEASE	70,926	15.71%
CODE OK	68,006	15.07%
BILLED	67,448	14.94%
CHANGE DIAGN	45,451	10.07%
DELETE	8,225	1.82%
REOPEN	4,669	1.03%
CODE NOK	3,620	0.80%
STORNO	2,973	0.66%
REJECT	2,016	0.45%
SET STATUS	705	0.16%
EMPTY	449	0.10%
MANUAL	372	0.08%
JOIN-PAT	358	0.08%
CODE ERROR	75	0.02%
CHANGE END	38	0.01%
ZDBC_BEHAN	1	0%

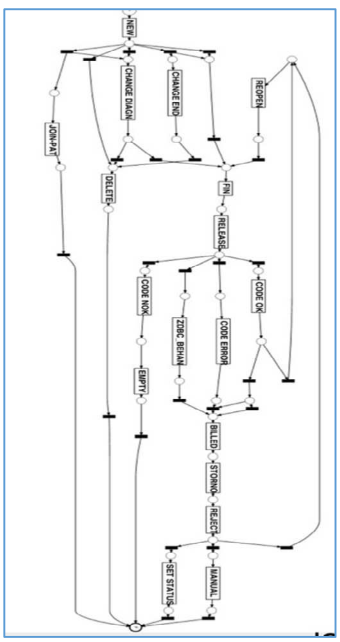

Fig 4. Control flow model of medical billing event log

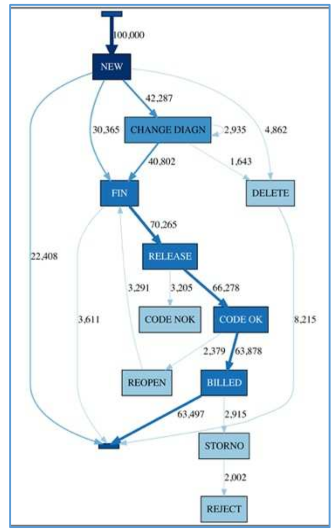

Fig 5. C-Net model of medical billing event log

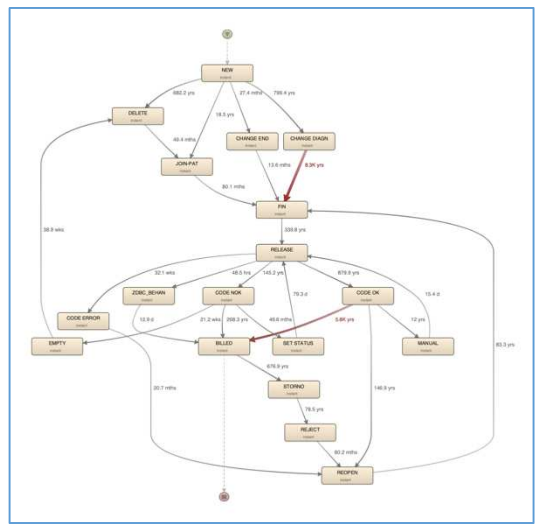

Fig 6. Time and Frequency perspective of event log

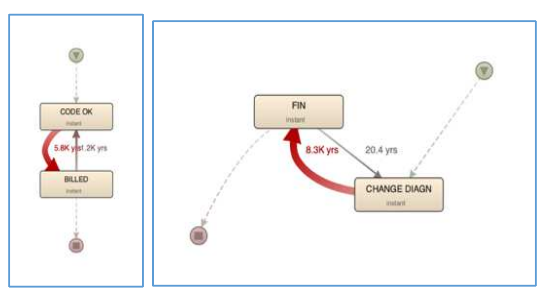

Fig. 1. Activities between CODE OK and BILLED FIN and CHANGE DIAGN

IV. CONCLUSION

As process mining and business process management research continue to grow in popularity, researchers need to apply the techniques to specific business domains. This paper introduce process thinking to medical billing process using the process mining L* life cycle model. We provided a brief overview of the challenges in the current medical billing process in the operations of the current medical billing process ERP systems, a comprehensive control flow of the medical billing process and analyzing of an event log extracted from a hospital billing ERP system to check activity flow, performance and exceptions between some activities. Activity flow in the event log was analyzed and explained using a control flow model and C-net generated using ProM6.10 and we established that of all the 100,000 of the event log which represent actual visit, only 63% got billed and about 3% of the billed got rejected by the payers. Time perspective was introduced using Disco fussy miner to detect highest activity duration between (CODE OK, BILLED, FIN and CHANGE DIAGNOSIS) which represent the actual diagnosis, superbill coding and billing of the visits. The event log used in this research does not capture the complete activity flow for a real-world medical billing system. Hence, we cannot directly compare the process flow map of the event log with the handmodel to give precise operational support. Future work on process mining for the medical billing process should focus on capturing event data for the entire medical billing process flow. The event log should include data from the providers' office, biller office, clearinghouse office and Payer's office.

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